Fundamentals / Foundation of Data Science

Data Science 9CFU

Computer Science 6CFU

Indro Spinelli

Sapienza University of Rome

About me

BSc Engineering in Computer Science
MSc Artificial Intelligence and Robotics
PhD in ICT (Neural Networks)
@ Sapienza University of Rome

Assistant Professor @ Sapienza University of Rome DI
Perception and Intelligence Lab (PINLab) Lead by Prof. Galasso

- AI and CV for Robotics
- Test the limits of in-context learning in LLMs
- AI Safety (Unlearning and Oversight)

TA: Simone Facchiano

- **Ph.D.** in Artificial Intelligence (currently)
  - *Under the supervision of Prof. Spinelli, Galasso and Rodolà*
  - *Generative AI, Manifold Learning, Representation Hypothesis*

- B.Sc. in **Statistics** *(Statistica Gestionale)* @ Sapienza

- M.Sc. in **Data Science** at Sapienza
  - Erasmus Exchange at Maastricht University
  - Honours Program
  - Master's Thesis at CINECA

TA: Leonardo Rocci

- B.Sc. in **Mathematics** @ SAPIENZA UNIVERSITÀ DI ROMA
- M.Sc. in **Data Science** (currently)
- My email: rocci.1922496@studenti.uniroma1.it

Lectures

Mondays 10:00-13:00 @ AULA 2L Via Castro Laurenziano 7a

Fridays 11:00-13:00 @ AULA 2L Via Castro Laurenziano 7a

Data Science studendents 9 CFU must follow Prof. Cinelli's Lectures on Tuesdays Aula Informatica 15 - Via Tiburtina 205

Office hours (by appointment)

Wednesday 13:30-14:30 in room 24, 2° floor, building G, via Regina Elena 295

Site and Groups

All communication will happen trough Goole Classroom:

Including slides, assignments and news. SUBSCRIBE now!

ya5p7ljo

The link can also be found on my personal page.

Exam

6 CFU Foundation of Data Science

- 50% Theory
- 50% Practice (of which 25% Challenge and 25% Final Project)

9 CFU Fundamentals of Data Science

- 34% Theory
- 33% Practice (of which 50% Assignments and 50% Final Project)
- 33% Prof. Cinelli's Theory part

Theory

A multiple choice exam lasting 30 minutes with 20 questions.

17. How is the reparametrization trick mathematically expressed in Variational Autoencoders?

- ✓ $z = \mu + \sigma \times \varepsilon$, where $\varepsilon \sim N(0,1)$
- $z = \mu + \log(\sigma^2) \times \varepsilon$, where $\varepsilon \sim N(0,1)$
- $z = \mu \times \sigma + \varepsilon$, where $\varepsilon \sim N(0,1)$
- $z = \exp(\mu + \sigma) \times \varepsilon$, where $\varepsilon \sim N(0,1)$

At least 1 correct, and max 0 wrong (1p) 0p 1 1p

Practice

The assignments and the final project must be sumbitted in groups before the written exam.

- Groups must be composed of 3 students (no more, no less)
- Build your team today!
- Compile this google form:

If you miss an assignment due to visa issues, you must still form a group. That missing assignment will be evaluated via an oral exam.

Calendar

Challenge: 15 Oct – 15 Nov

Final Project: 15 Nov – 15 Dec

- Project Announcement: 10 Nov
- Presentation of Ideas: 25 Nov
- Project Presentation: Last week of lessons in Dec

Final report & code must be submitted by 23 Dec.

Evaluation criteria

Presentation skills [time, pace, tone, energy, interaction, ...]

Language

Slides [style, figures, citations correctness, ...]

Completeness [intro, motivation, method description, experimental validation, conclusion, demo, ...]

Technical Quality / Contribution

Demo / Example

Group Work / Organization

Q&A

Plagiarism

Plagiarism is an Act of Fraud

Plagiarism is strictly prohibited and constitutes a severe violation of ethical standards. This includes submitting any assignment or project—whether source code or written report—that strongly resembles someone else's work, including other student submissions or online resources. All submitted projects will be screened for plagiarism using automatic detection tools. Any work found to be plagiarized will be invalidated.

Use of Generative AI

- **Success Without AI:** You do not need AI tools to succeed. Exams are completed **without AI access**.
- **Learning Aid:** AI tools may be used to **understand and explore concepts**, but **all use must be documented**.
- **Academic Integrity:** AI outputs can be incorrect. **Directly submitting AI-generated work** is a violation.
- **Homework:** You must **write your own solutions**; AI should **assist learning, not replace your work**.

Support and Counseling

Sapienza offers dedicated support for students with disabilities and specific learning disorders (DSA). For assistance reach out to:

sportellodisabili@uniroma1.it or counselingdsa@uniroma1.it

Or to the faculty point of contact:

Prof. Tiziana Calamoneri
Coordinator for Disabilities and DSA, I3S Faculty
http://wwwusers.di.uniroma1.it/~calamo/

Prerequisites

Calculus and Linear Algebra

taking derivatives, understanding matrix vector products

Basic Probability and Statistics

basics of probabilities, gaussian distributions, mean, standard deviation, etc

3Blue1Brown's Youtube series on the Essence of Linear Algebra

Mathematics for Machine Learning (https://mml-book.github.io/book/mml-book.pdf)

Chapters: 2, 3.1-3.7, 4.1-4.6 ,5.1, 6.1-6.5

Syllabus

Intro Dataset Visualization

Linear Algebra SVD

Regression Bias Variance Regularization Interpretability

Classification Evaluation Calibration

KNN Decision Trees Random Forest Boosting

MLP Backpropagation

Image Representation Convolution

CNN Dropout Normalisation Finetuning

Text Representation Attention Transformer

PCA AutoEncoders Variational AE Multimodal Learning

Books and Material

Deisenroth, Faisal, Ong. Mathematics for Machine Learning. Cambridge Press, 2020.

S. Prince. *Understanding Deep Learning.* MIT Press, 2023.

Watt, Borhani, Katsaggelos, Machine Learning Refined

Both freely available!

All the slides and other materials will be published in the google classroom. Hopefully before, most likely after the lesson.

What is Data Science?

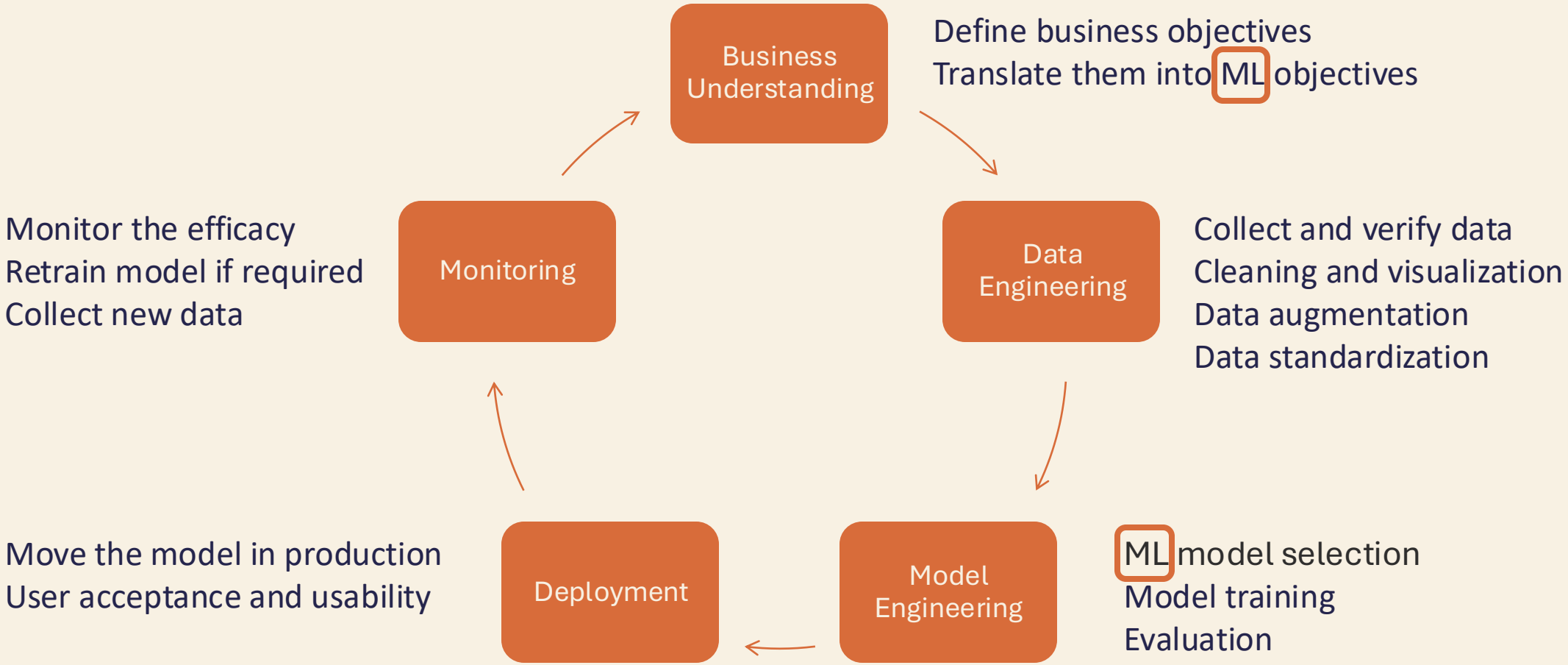
And why we focus on machine learning

Definition

"Data science is an interdisciplinary field that focuses on extracting actionable insights and knowledge from data. It involves the collection, cleaning, management, and analysis of both structured and unstructured data, using techniques rooted in statistics, mathematics, computer science, and domain-specific expertise."

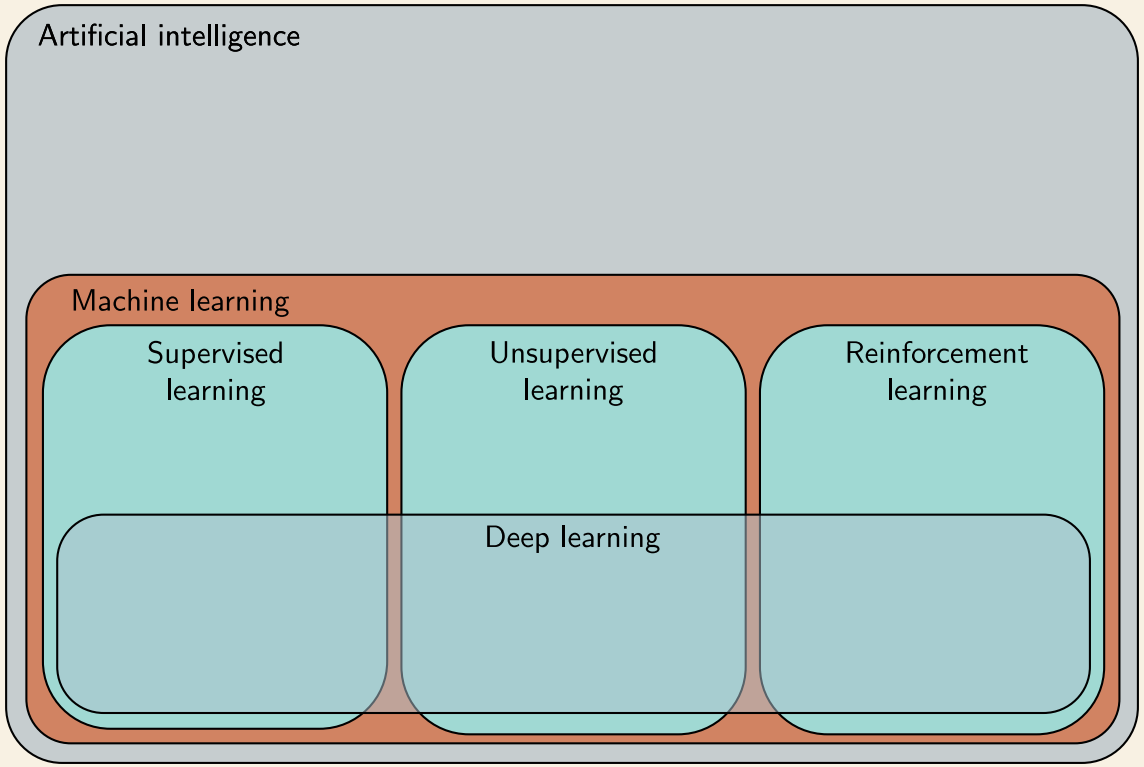
ChatGPT

Data Science Lifecycle



What is ML (Machine Learning)

Artificial Intelligence is the field of computer science dedicated to creating agents capable of performing tasks that typically require human intelligence such as learning reasoning and problem solving.



Machine Learning

Definition by Tom Mitchell (1998)

Machine Learning is the study of algorithms that:

- improve their performance P
- at some task T
- with experience E.

A well-defined learning task is given by <P, T, E>.

Examples

T: Playing checkers
P: Percentage of games won against an arbitrary opponent
E: Playing practice games against itself
T: Recognizing hand-written words
P: Percentage of words correctly classified
E: Database of human-labeled images of handwritten words
T: Driving on four-lane highways using vision sensors
P: Average distance traveled before a human-judged error
E: A sequence of images and steering commands recorded while observing a human driver.
T: Categorize email messages as spam or legitimate.
P: Percentage of email messages correctly classified.
E: Database of emails, some with human-given labels

Why machine learning?

Challenge: Many tasks are too complex to hand-code

- Recognizing people and objects
- Understanding human speech

Hard to design explicit rules (vision, speech, language)

Machine Learning Approach: Build algorithms that **learn automatically** from data or experience

ML vs Statistics

Similarities

- Both uncover patterns in data
- Both use calculus, probability, and linear algebra
- Many shared algorithms

Differences

- **Statistics**: focus on inference, conclusions, interpretability, rigor
- **Machine Learning**: focus on prediction, scalability, autonomy

Takeaway

Same mathematics, different goals & emphasis

Machine learning

Supervised learning:

have labeled examples of the correct behavior

Unsupervised learning:

no labeled examples – instead, looking for “interesting” patterns in the data

Reinforcement learning:

learning system (agent) interacts with the world and learns to act rationally maximizing a scalar reward signal.

I/O

Supervised learning

Input: Examples of inputs and desired outputs

Output: Model that predicts output given a new input

Unsupervised learning

Input: Examples of some data (no “outputs”)

Output: Representation of structure in the data

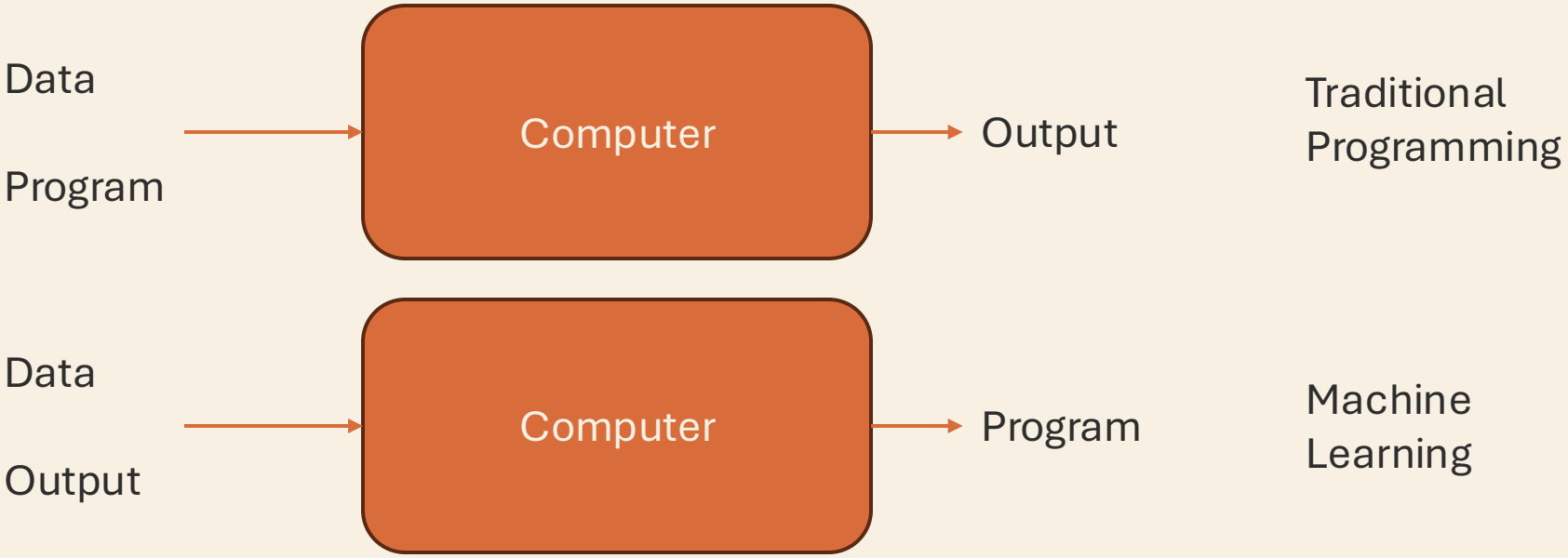
Reinforcement learning

Input: Sequence of interactions with an environment

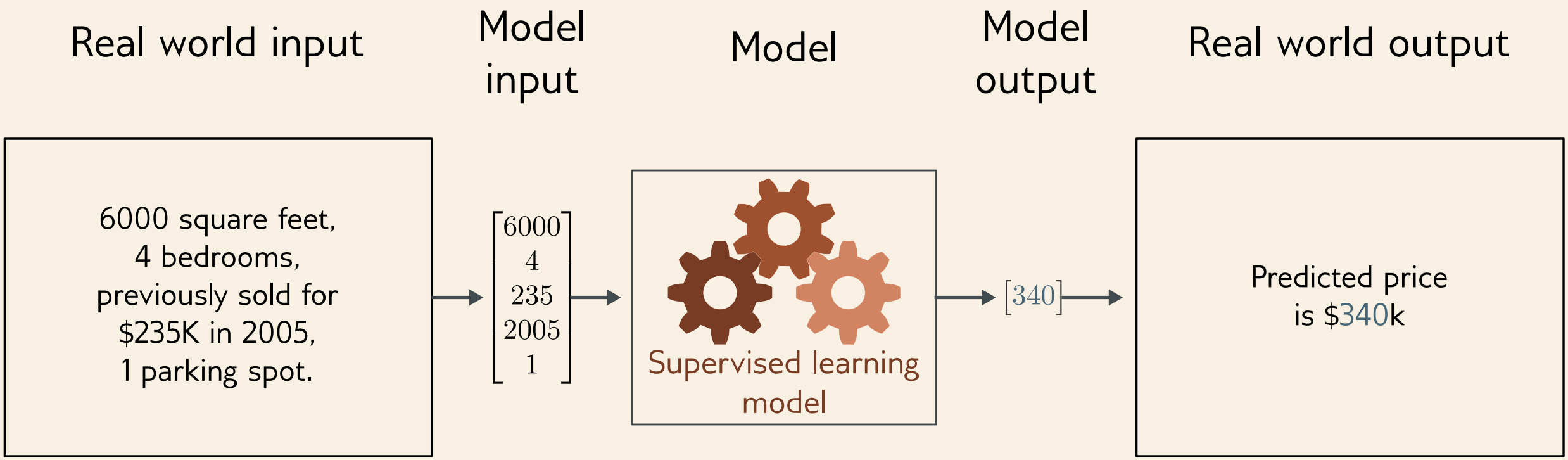
Output: Policy that performs a desired task

Supervised learning

- Define a mapping from input to output
- Learn this mapping from paired input/output data examples

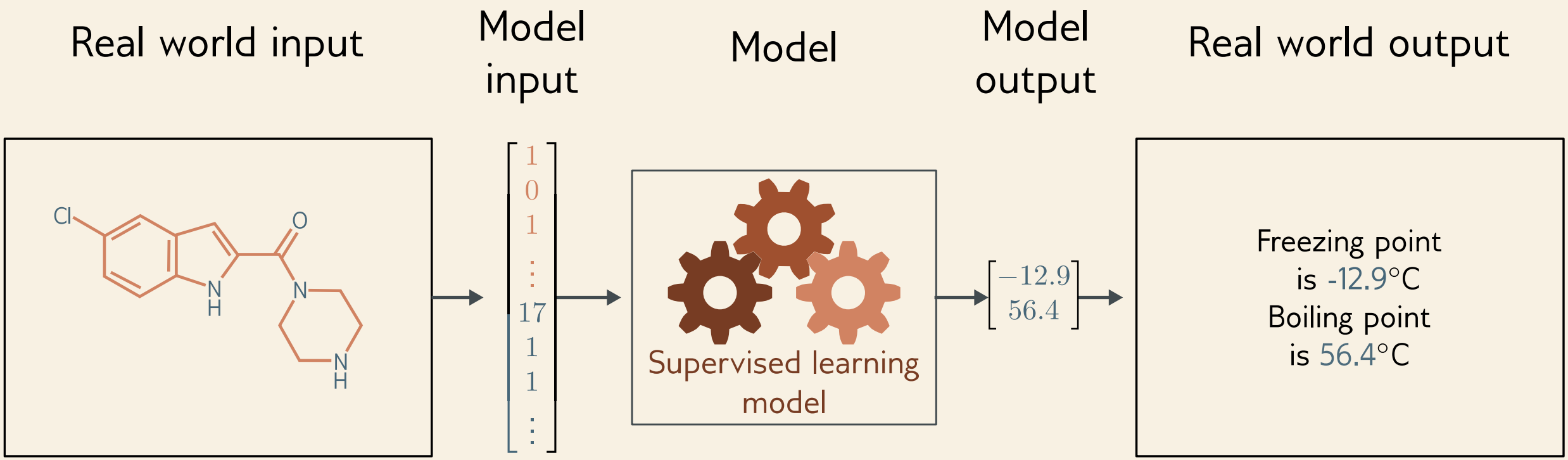


Regression



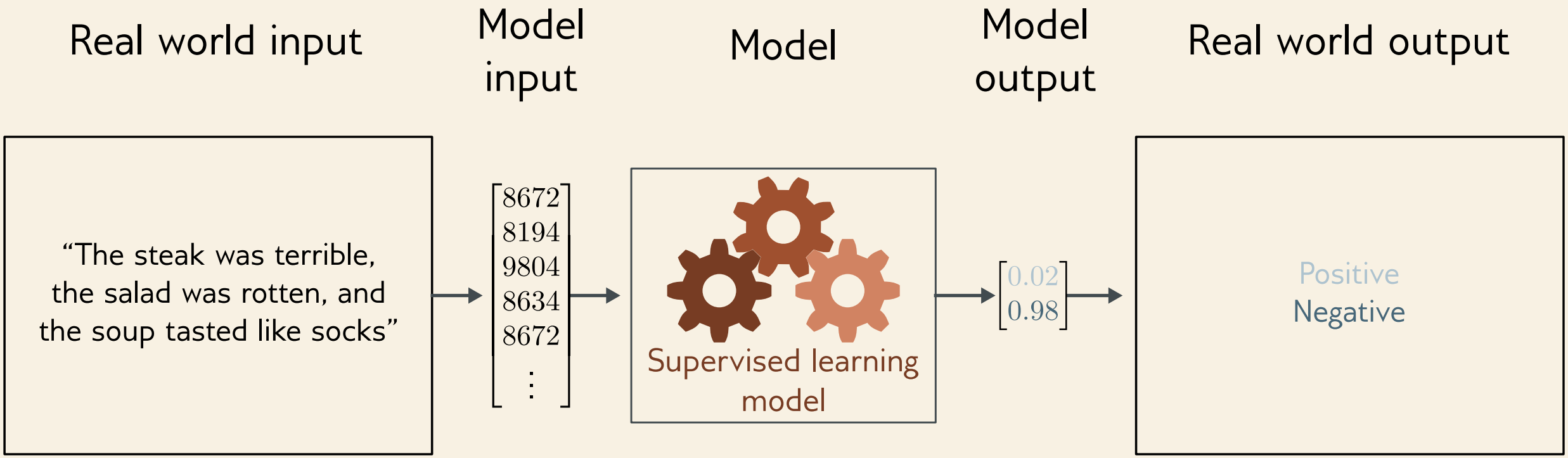
- Univariate regression problem (one output, real value)
- Fully connected network

Graph regression



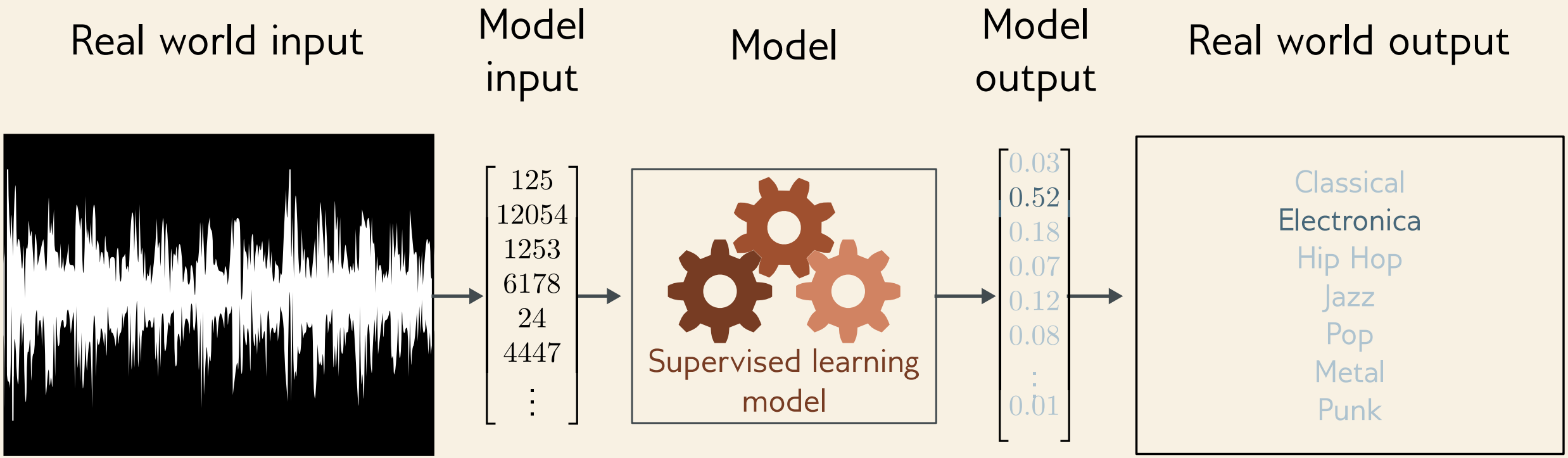
- Multivariate regression problem (>1 output, real value)
- Graph neural network

Text classification



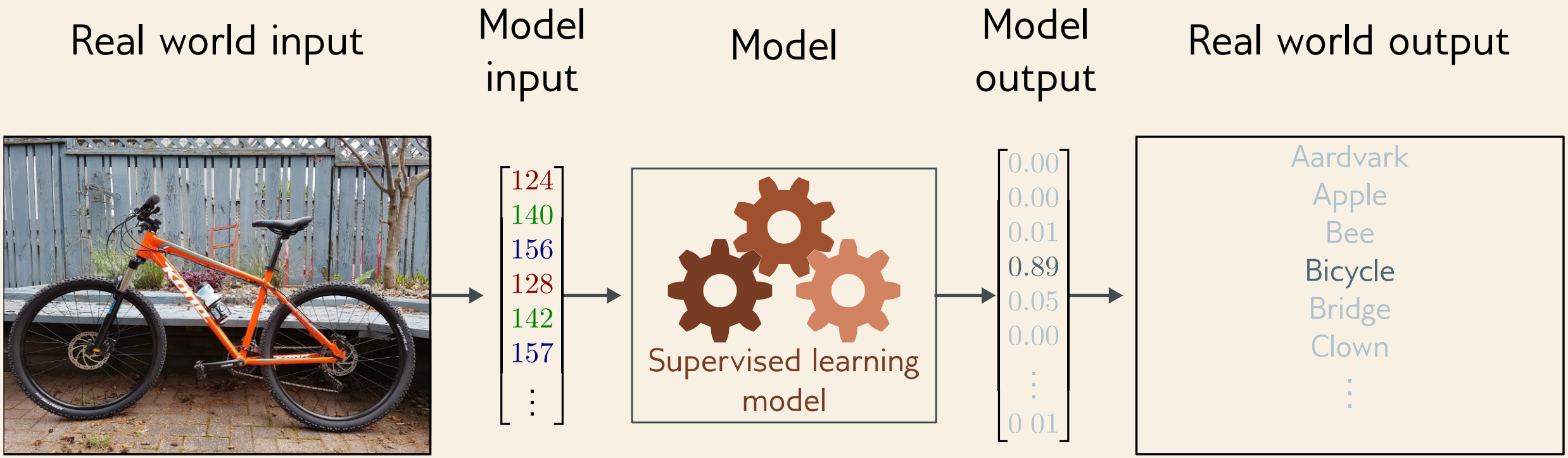
- Binary classification problem (two discrete classes)
- Transformer network

Music genre classification



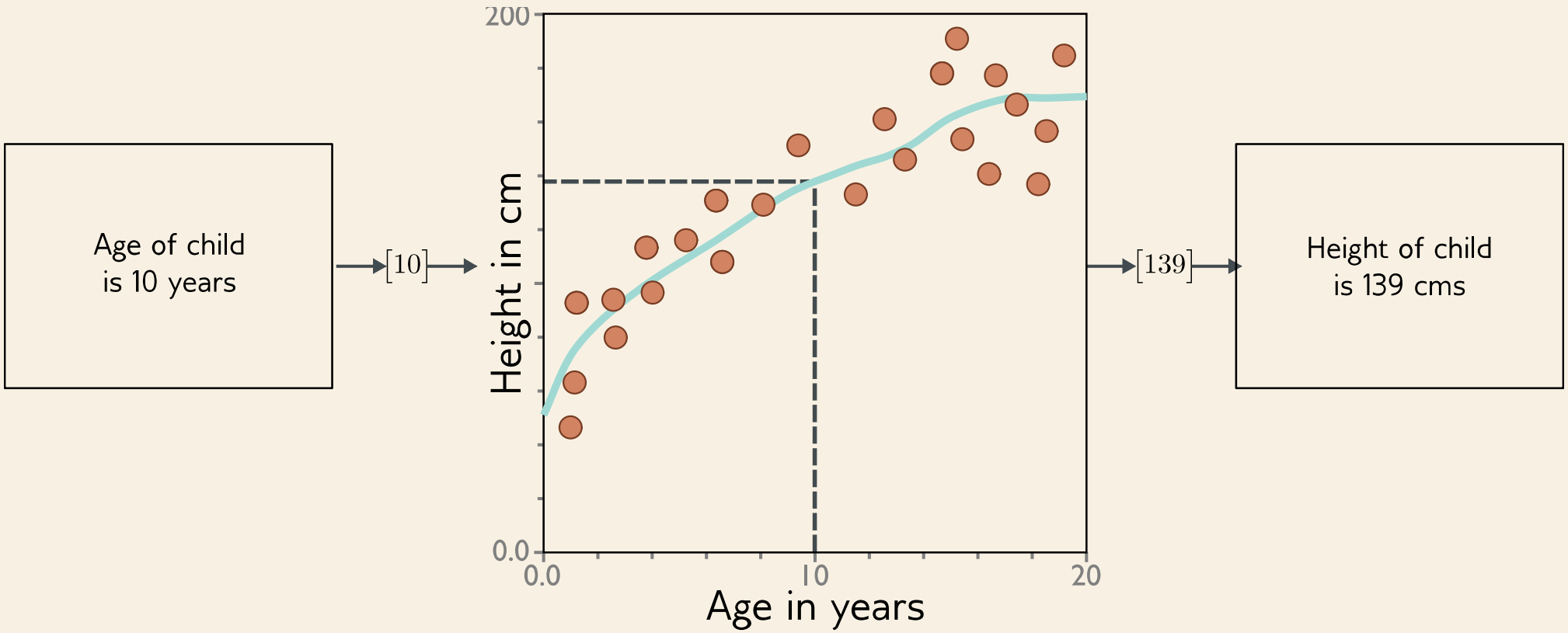
- Multiclass classification problem (discrete classes, >2 possible values)
- Recurrent neural network (RNN)

Image classification



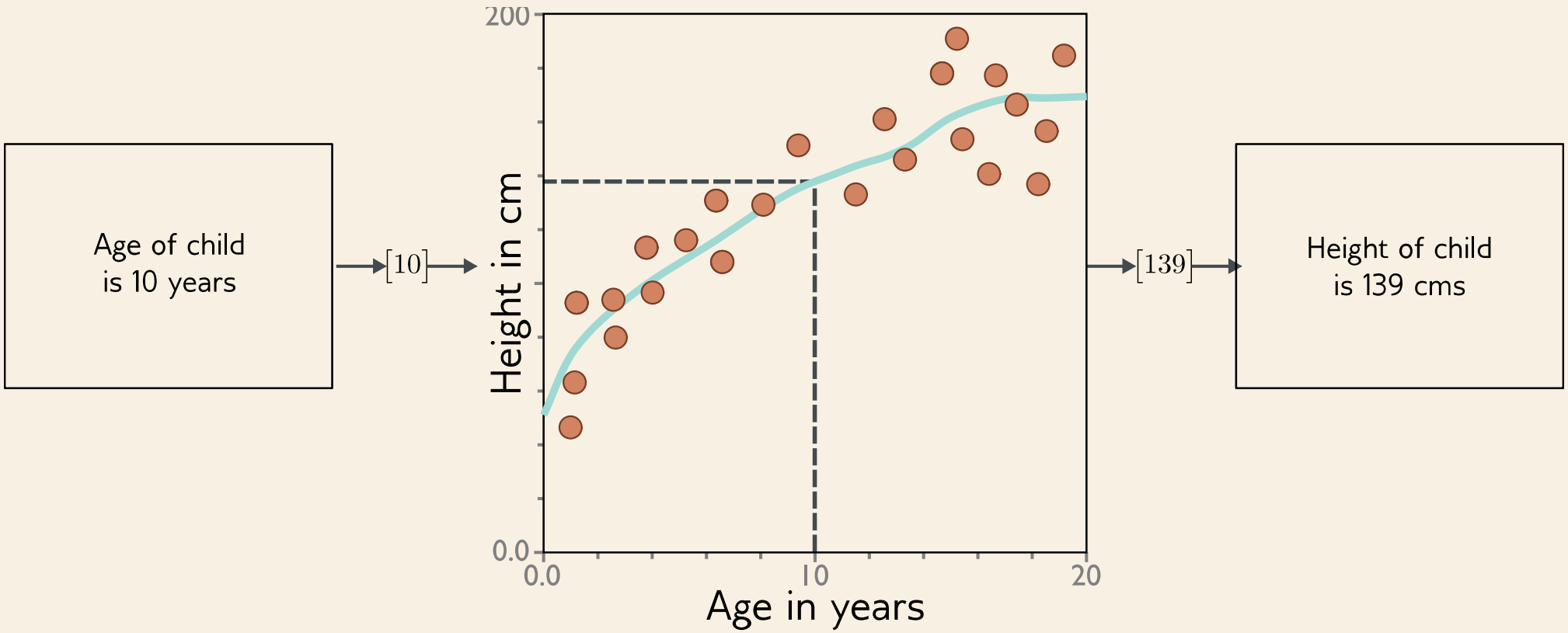
- Multiclass classification problem (discrete classes, >2 possible classes)
- Convolutional network

What is a supervised learning model?



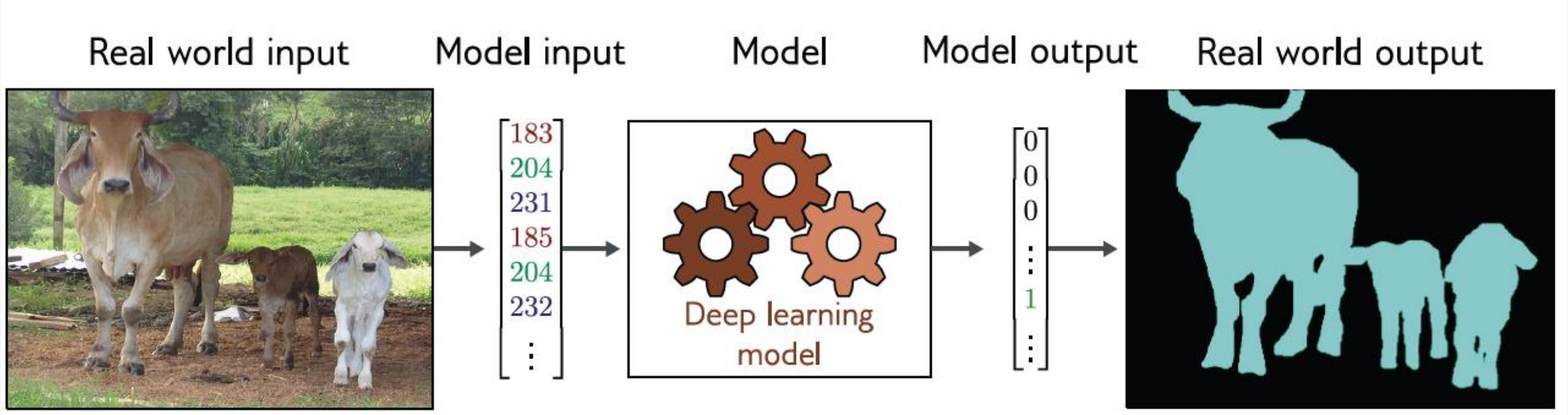
- An equation relating input (age) to output (height)
- Search through family of possible equations to find one that fits training data well

What is a supervised learning model?



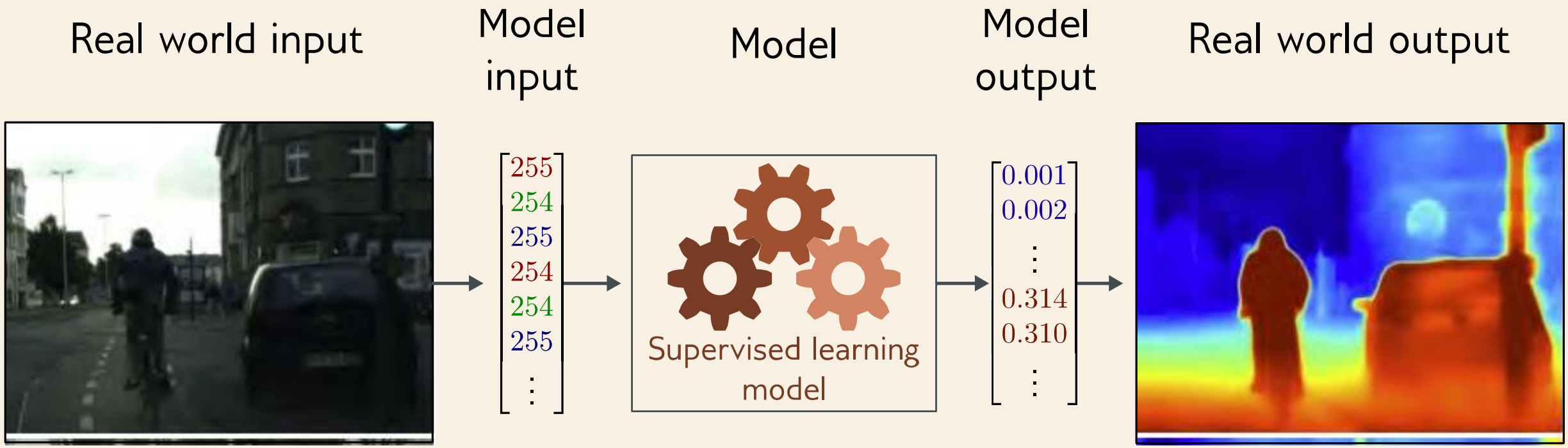
- Deep neural networks are just a very flexible family of equations
- Fitting deep neural networks = “Deep Learning”

Image segmentation



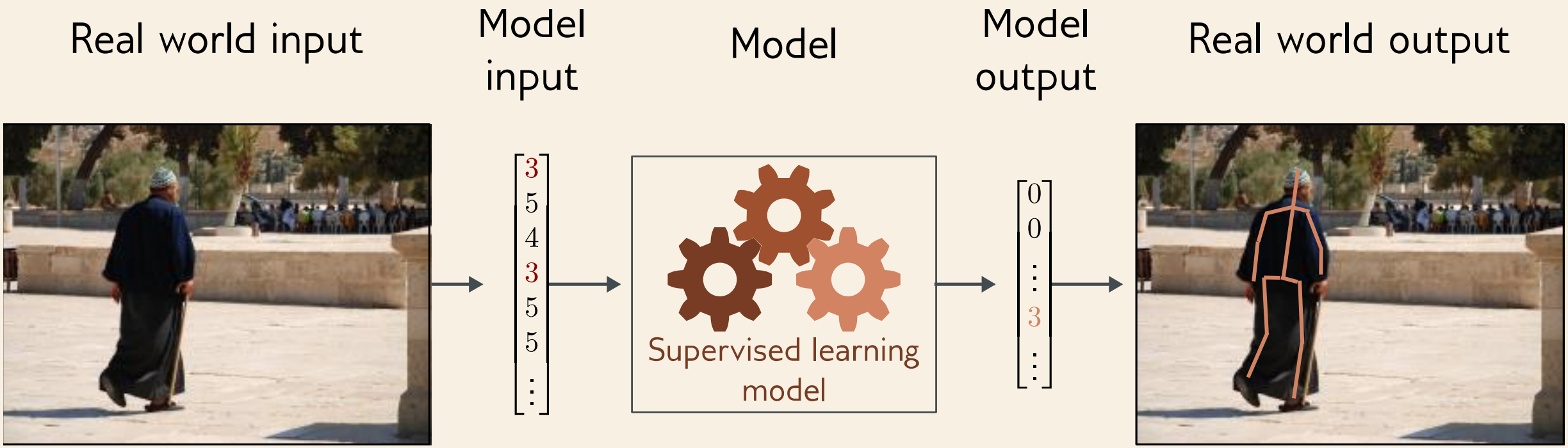
- Multivariate binary classification problem (many outputs, two discrete classes)
- Convolutional encoder-decoder network

Depth estimation



- Multivariate regression problem (many outputs, continuous)
- Convolutional encoder-decoder network

Pose estimation



- Multivariate regression problem (many outputs, continuous)
- Convolutional encoder-decoder network

Translation

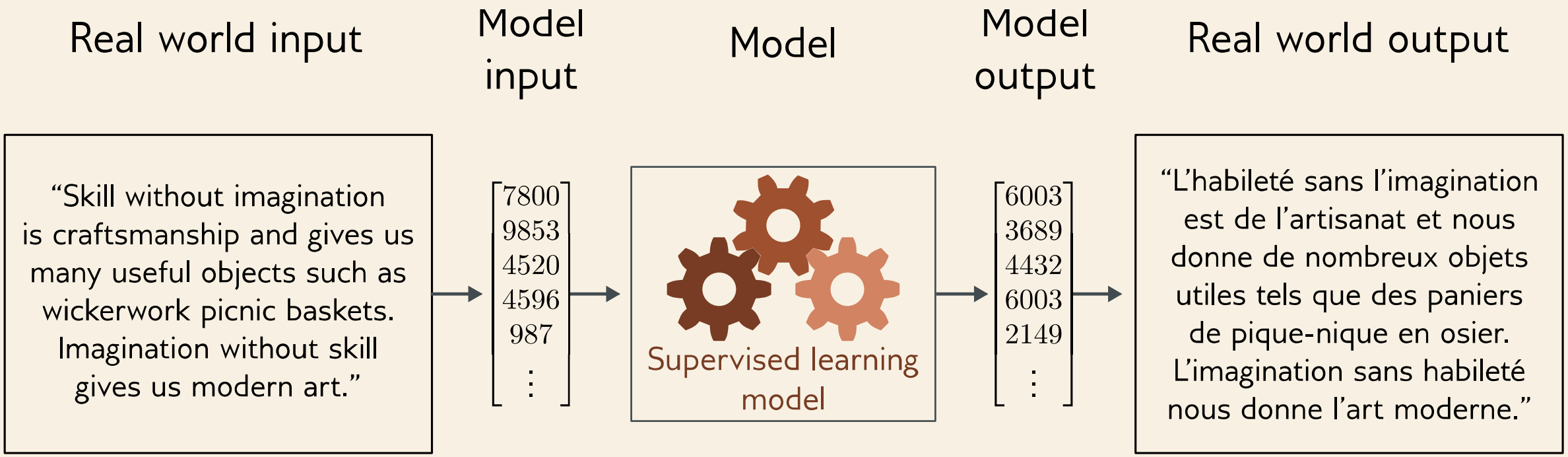


Image captioning

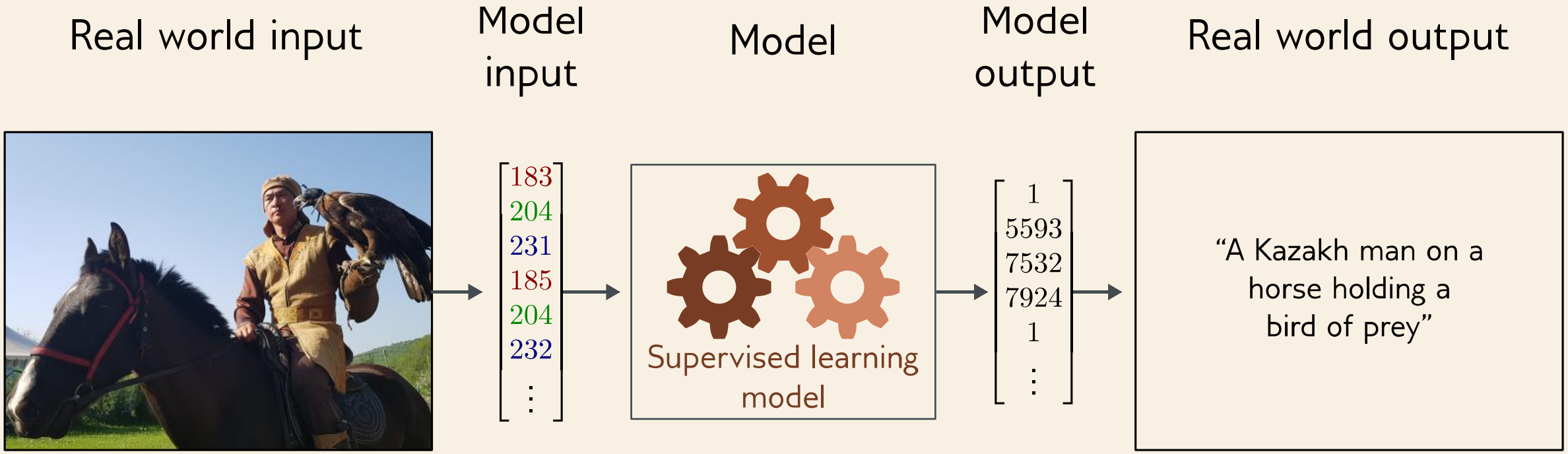
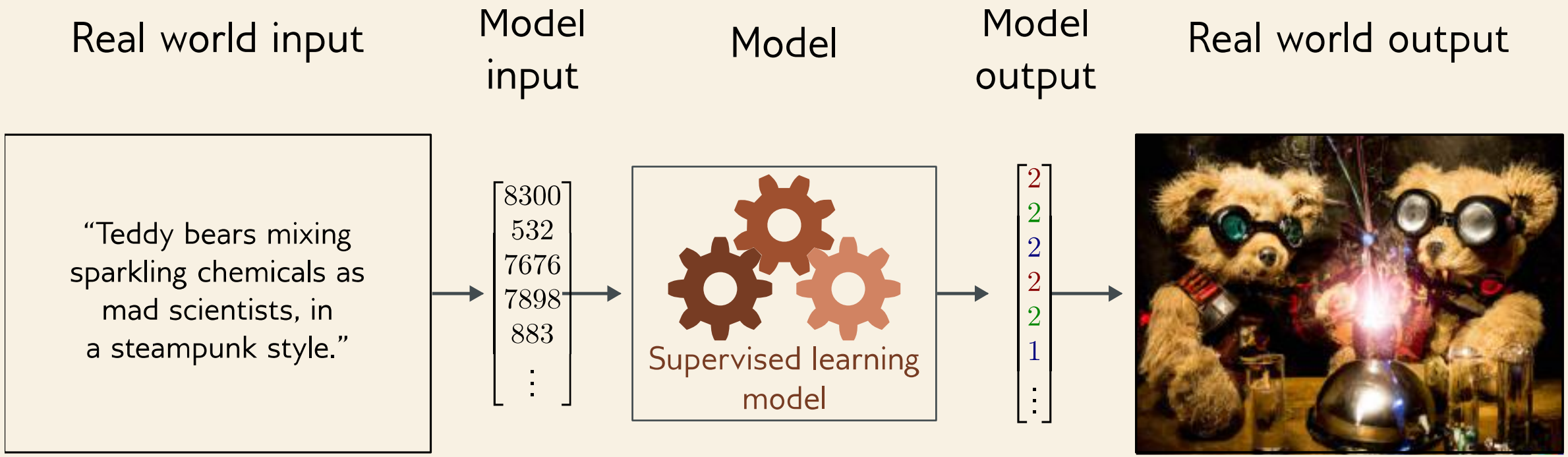


Image generation from text



What do these examples have in common?

- Very complex relationship between input and output
- Sometimes may be many possible valid answers
- But outputs (and sometimes inputs) obey rules

"A Kazakh man on a horse holding a bird of prey"

Language obeys grammatical rules

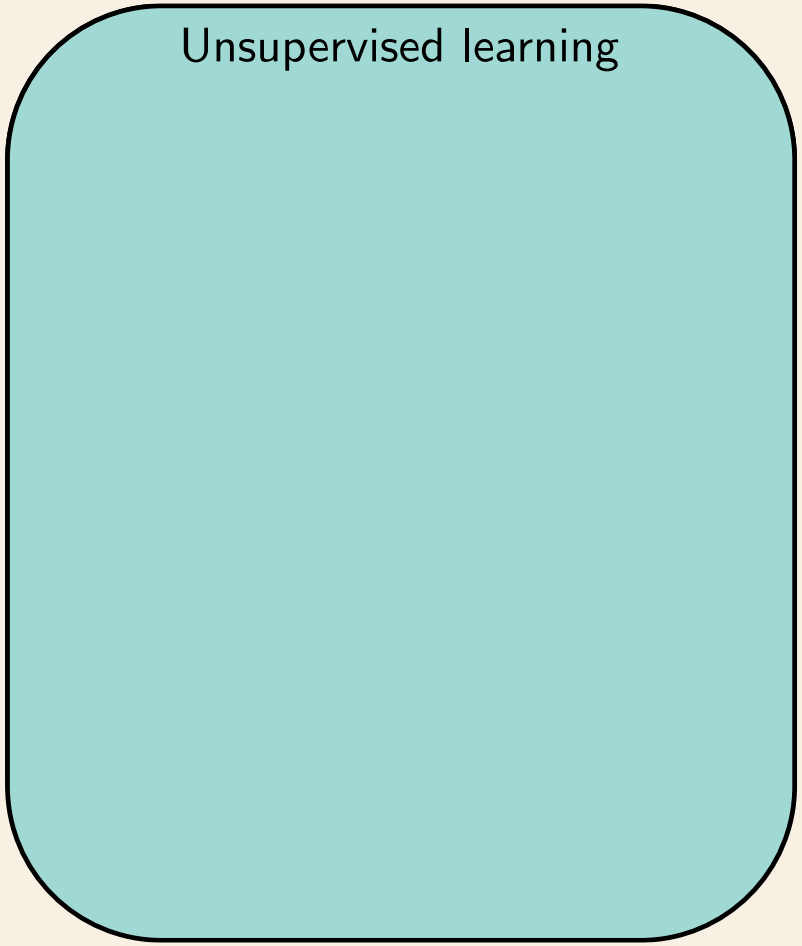
Natural images also have "rules"

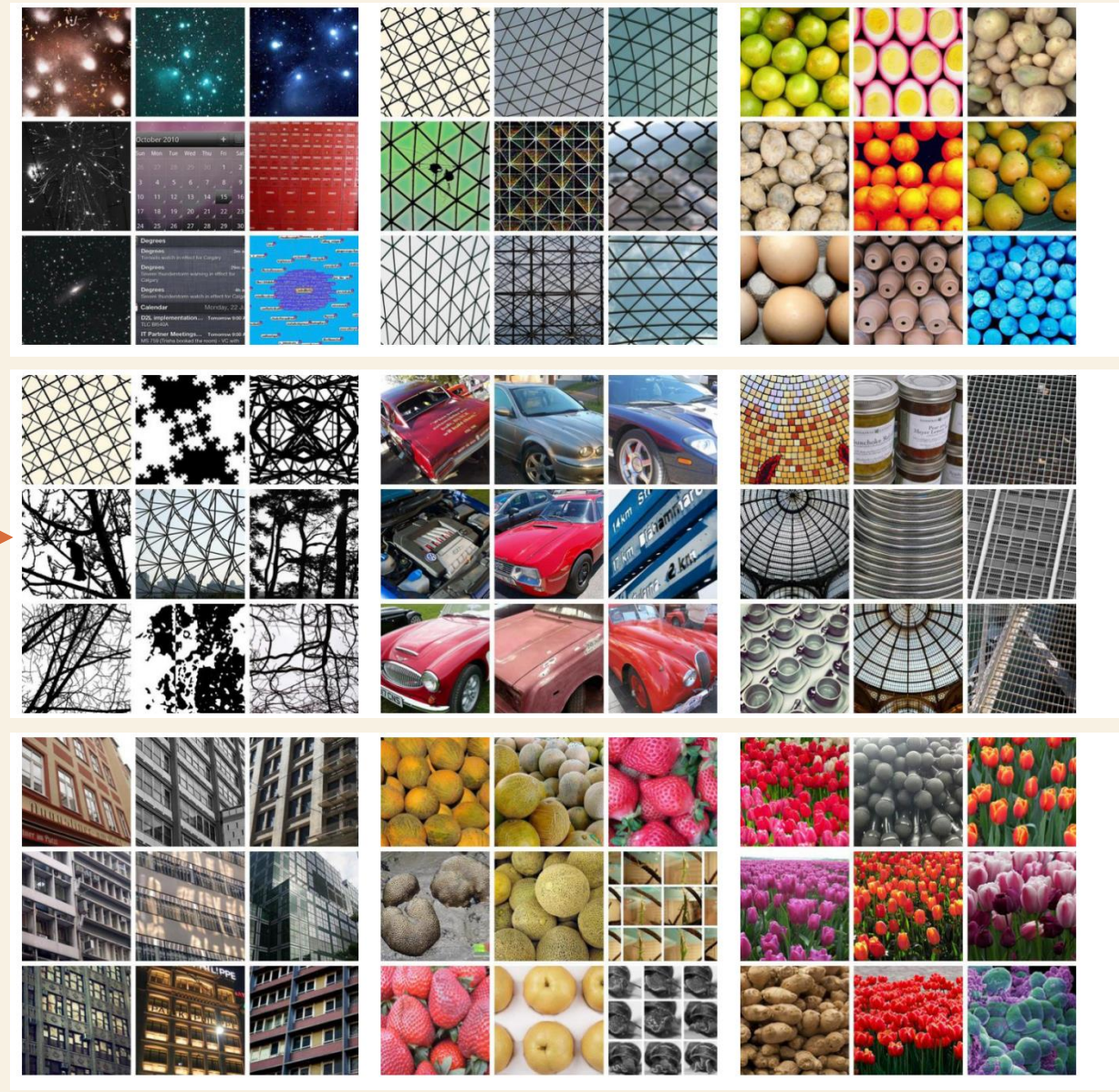
Idea

- Learn the “grammar” of the data from unlabeled examples
- Can use a gargantuan amount of data to do this (as unlabeled)
- Make the supervised learning task easier by having a lot of knowledge of possible outputs

Unsupervised Learning

- Learning about a dataset without labels
  - Clustering
  - Finding outliers
  - Generating new examples
  - Filling in missing data

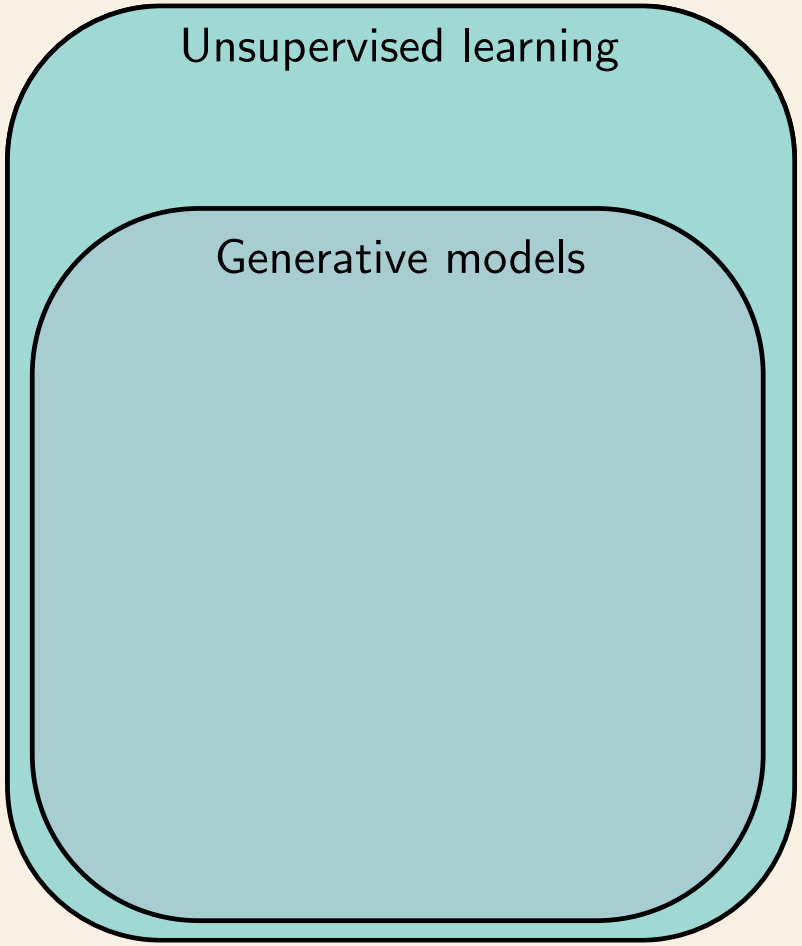




DeepCluster: Deep Clustering for Unsupervised Learning of Visual Features (Caron et al., 2018)

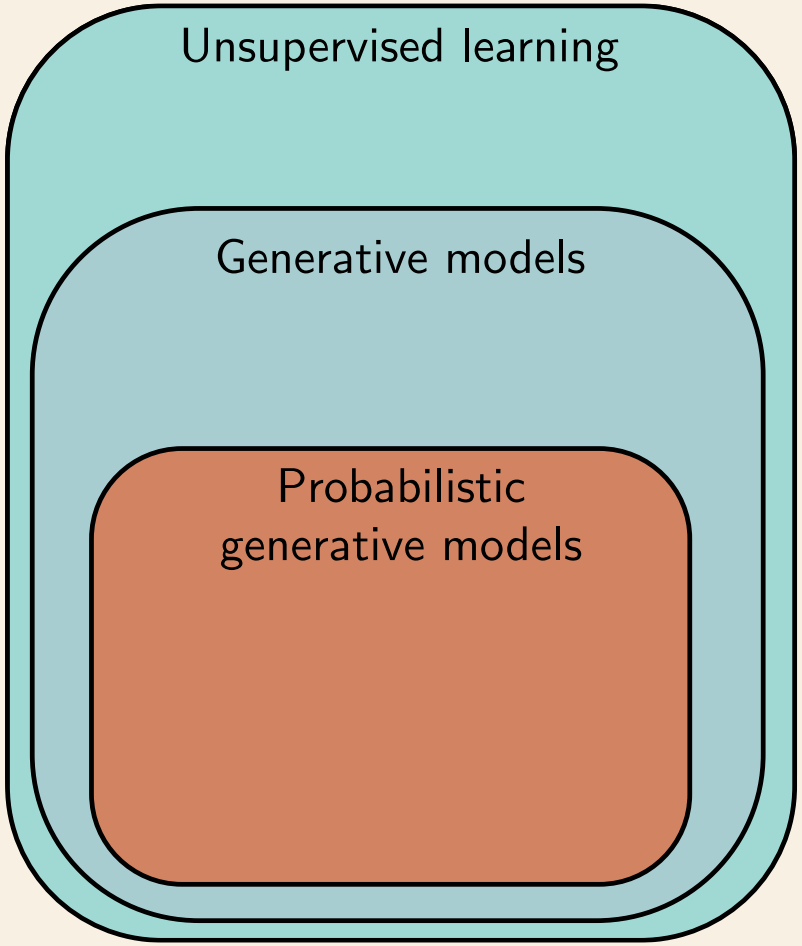
Unsupervised Learning

- Learning about a dataset without labels
  - e.g., clustering
- Generative models can create examples
  - e.g., generative adversarial networks

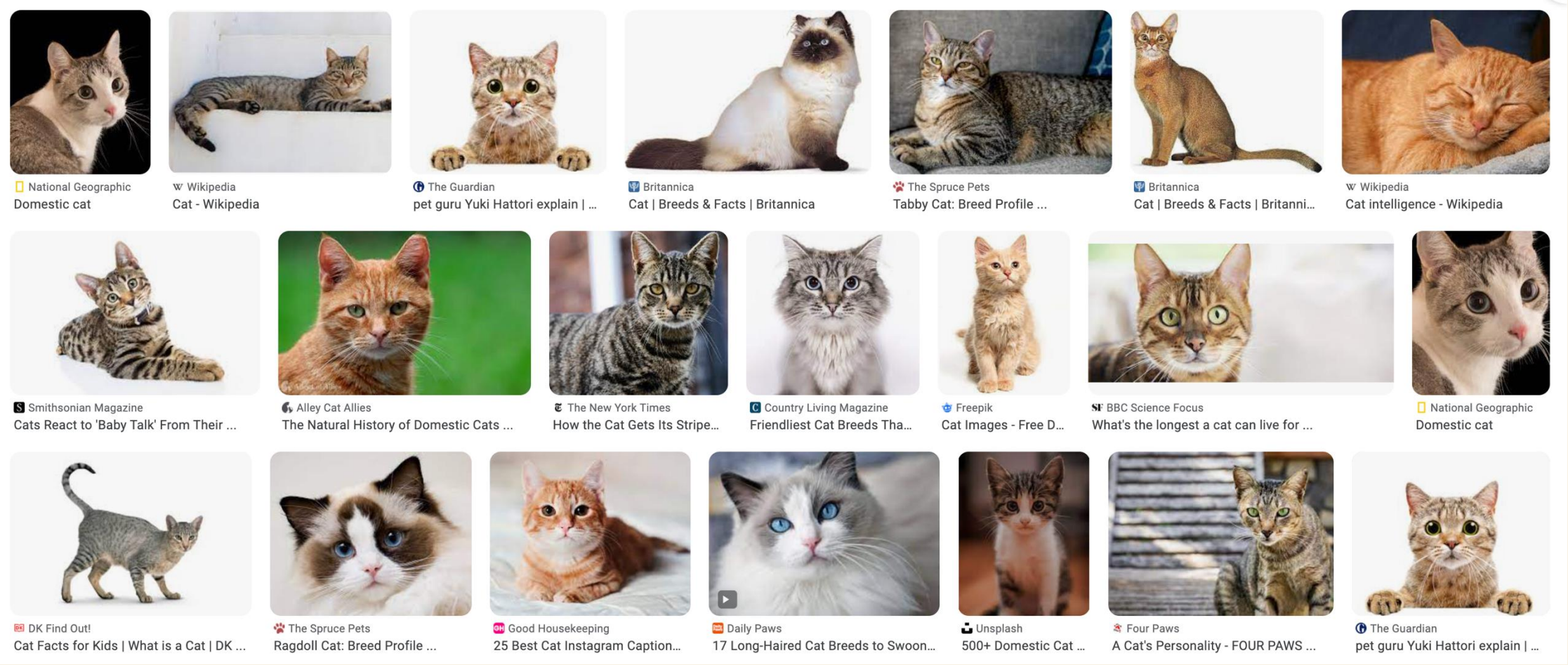


Unsupervised Learning

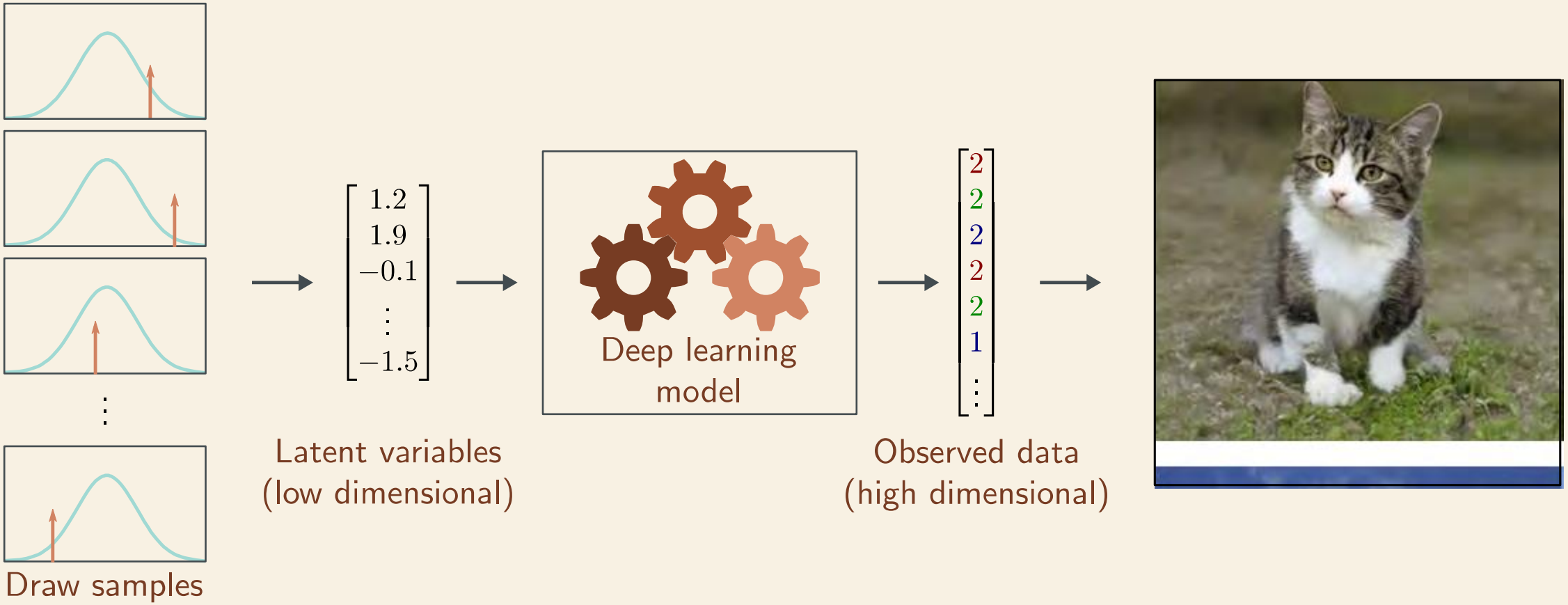
- Learning about a dataset without labels
  - e.g., clustering
- Generative models can create examples
  - e.g., generative adversarial networks
- PGMs learn distribution over data
  - e.g., variational autoencoders,
  - e.g., normalizing flows,
  - e.g., diffusion models



Generative models



Latent variables



Reinforcement learning

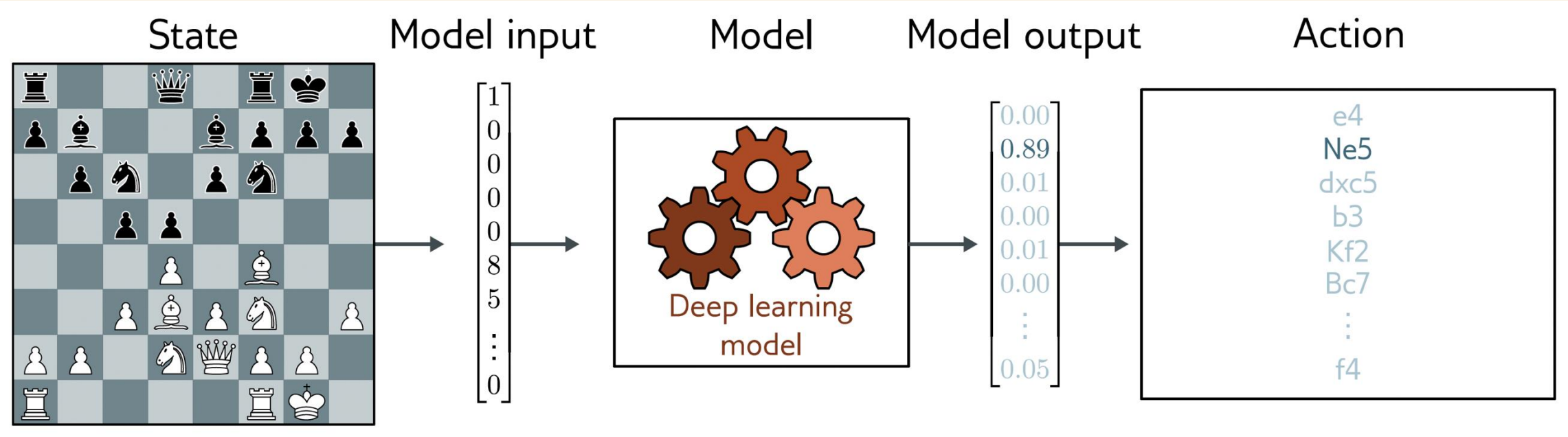
- A set of states
- A set of actions
- A set of rewards

- Goal: take actions to change the state so that you receive rewards

- You don’t receive any data – you have to explore the environment yourself to gather data as you go

Example: chess

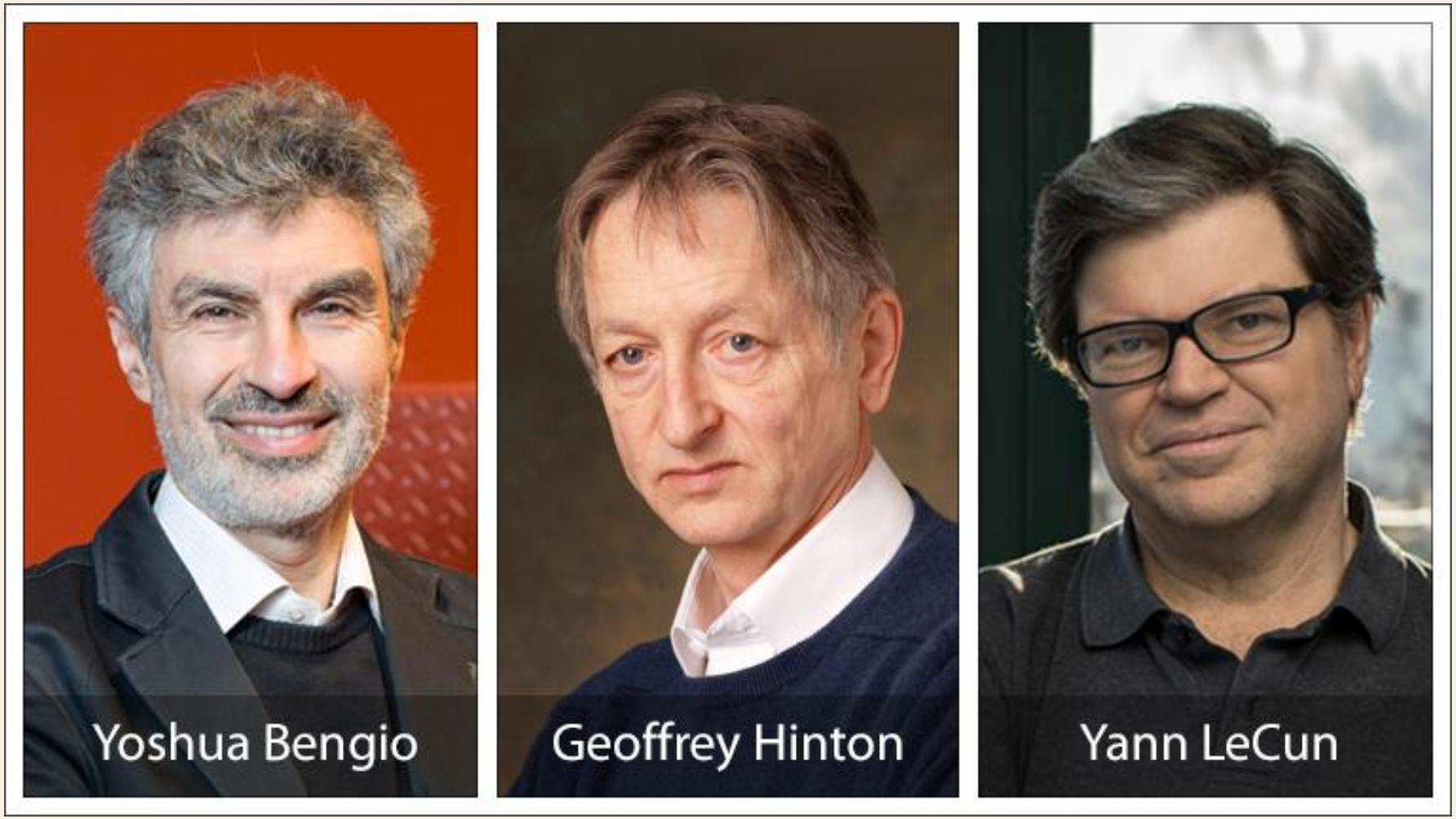
- States are valid states of the chess board
- Actions at a given time are valid possible moves
- Positive rewards for taking pieces, negative rewards for losing them



Why is this difficult?

- Stochastic
  - Make the same move twice, the opponent might not do the same thing
  - Rewards also stochastic (opponent does or doesn’t take your piece)
- Temporal credit assignment problem
  - Did we get the reward because of this move? Or because we made good tactical decisions somewhere in the past?
- Exploration-exploitation trade-off
  - If we found a good opening, should we use this?
  - Or should we try other things, hoping for something better?

2018 Turing award winners



Deep Learning Landmarks

- 1958 Perceptron (Simple neural model)
- 1986 Backpropagation (Practical Deep Neural networks)
- 1989 Convolutional networks (Supervised learning)
- 2012 AlexNet Image classification (Supervised learning)
- 2014 Generative adversarial networks (Unsupervised learning)
- 2014 Deep Q-Learning -- Atari games (Reinforcement learning)
- 2016 AlphaGo (Reinforcement learning)
- 2017 Machine translation (Supervised learning)
- 2019 Language models ((Un)supervised learning)
- 2022 Dall-E2 Image synthesis from text prompts ((Un)supervised learning)
- 2022 ChatGPT ((Un)supervised learning)
- 2023 GPT4 Multimodal model ((Un)supervised learning)

Slides based on the introduction by *Understanding Deep Learning.*